Simulation Report

Experiment 5 - Hardware Q-Learning Agent on the PYNQ-Z2

1. Setup

Item	Value
Simulator	Icarus Verilog 12 (iverilog -g2012, vvp)
Design under test	rl_agent_top (RTL/*.sv, 10 modules)
Testbench	Testbench/rl_agent_tb.sv
Waveform dump	Simulation/wave_dump.sv records selected signals to rl_wave.vcd: the start of training and the last episode
Clock	50 MHz (20 ns period, the same as FCLK0 on the board)
Commands	iverilog -g2012 -o rl_sim.vvp -s rl_agent_tb -s wave_dump ../RTL/*.sv ../Testbench/rl_agent_tb.sv wave_dump.sv; vvp rl_sim.vvp; python plot_waveform.py

2. What the testbench checks

The testbench plays the part of the ARM processor. It drives the AXI4-Lite bus of rl_agent_top, reads the default settings, starts training on the maze built into the hardware and polls the statistics registers until the done flag is set. Then it reads the results back:

#	Check	Pass condition
1	Default settings	ALPHA reads 0x1A, GAMMA reads 0xF3 and MAX_EPISODES reads 2000
2	Training runs to the end	the episode counter reaches 2000 and at least one goal is reached
3	Shortest path found	the best path register reads 14 steps, the shortest path through the maze
4	Q-table learned	Q((7,6), RIGHT), the move into the goal, is above 80.0; the episode history buffer is not empty

Any wrong value stops the simulation with *$fatal*. The final PASS line means every check passed.

3. Result

PASS – all four checks passed. The agent reached the goal for the first time between episodes 358 and 420, and ended with 1335 goals in 2000 episodes and a best path of 14 steps. Q((7,6), RIGHT) = 96.37, so the Bellman update gave the move into the goal a high value. Training took about 4.4 ms of simulated time at 50 MHz.

Transcript

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Experiment 5 - Hardware Q-Learning Agent (PYNQ-Z2)
Simulation transcript, 2026-09-18 19:31
Simulator: Icarus Verilog version 12.0 (devel) (s20150603-1539-g2693dd32b)

> iverilog -g2012 -o rl_sim.vvp -s rl_agent_tb -s wave_dump ../RTL/*.sv ../Testbench/rl_agent_tb.sv wave_dum
p.sv   (run inside Simulation/)
../RTL/bellman_update.sv:121: sorry: constant selects in always_* processes are not currently supported (all
bits will be included).
../RTL/rl_axi_wrapper.sv:177: sorry: constant selects in always_* processes are not currently supported (all
bits will be included).
../RTL/rl_axi_wrapper.sv:177: sorry: constant selects in always_* processes are not currently supported (all
bits will be included).
../RTL/rl_axi_wrapper.sv:177: sorry: constant selects in always_* processes are not currently supported (all
bits will be included).
../RTL/rl_axi_wrapper.sv:177: sorry: constant selects in always_* processes are not currently supported (all
bits will be included).

> vvp rl_sim.vvp
VCD info: dumpfile rl_wave.vcd opened for output.
[PASS] default settings: alpha 0x1A, gamma 0xF3, 2000 episodes
training on the maze built into the hardware...
  episode 163, 0 goals reached
  episode 282, 0 goals reached
  episode 358, 0 goals reached
  episode 420, 9 goals reached
  episode 496, 60 goals reached
  episode 589, 139 goals reached
  episode 678, 214 goals reached
  episode 768, 294 goals reached
  episode 860, 372 goals reached
  episode 952, 449 goals reached
  episode 1049, 525 goals reached
  episode 1142, 605 goals reached
  episode 1236, 679 goals reached
  episode 1328, 756 goals reached
  episode 1422, 836 goals reached
  episode 1512, 916 goals reached
  episode 1605, 995 goals reached
  episode 1700, 1072 goals reached
  episode 1791, 1151 goals reached
  episode 1882, 1231 goals reached
  episode 1975, 1312 goals reached
  episode 2000, 1335 goals reached
finished: 2000 episodes, 1335 goals, best path 14 steps
[PASS] the agent found the shortest path through the maze
Q((7,6), RIGHT) = 96.37
[PASS] the value of stepping into the goal was learned
history entry 0: goal 1, 14 steps
PASS: the hardware agent learned to solve the maze
../Testbench/rl_agent_tb.sv:119: $finish called at 4407770000 (1ps)
```

The "sorry: constant selects" lines are Icarus warnings about a language feature it does not fully support; the always_comb block is simply re-evaluated when any bit of the vector changes, which gives the same result.

4. Waveform

Panel (a) shows the AXI write of CTRL = 1 that starts training and the first learning steps: the state machine goes through READ_Q (4), EXECUTE (5), READ_NEXT_Q (6), BELLMAN_WAIT (7), WRITE_Q (8) and CHECK_DONE (9) for every step, and with epsilon still high most actions are random.

Panel (b) shows the whole last episode: epsilon is at its minimum, every action is the learned best one, and the agent walks from (0,0) to (7,7) in 14 steps; then goal_pulse fires, the episode count reaches 2000 and done goes high. Panel (c) zooms in on the last step into the goal, where the +100 reward (0x6400 in 8.8 fixed point) raises the Q value of that move.

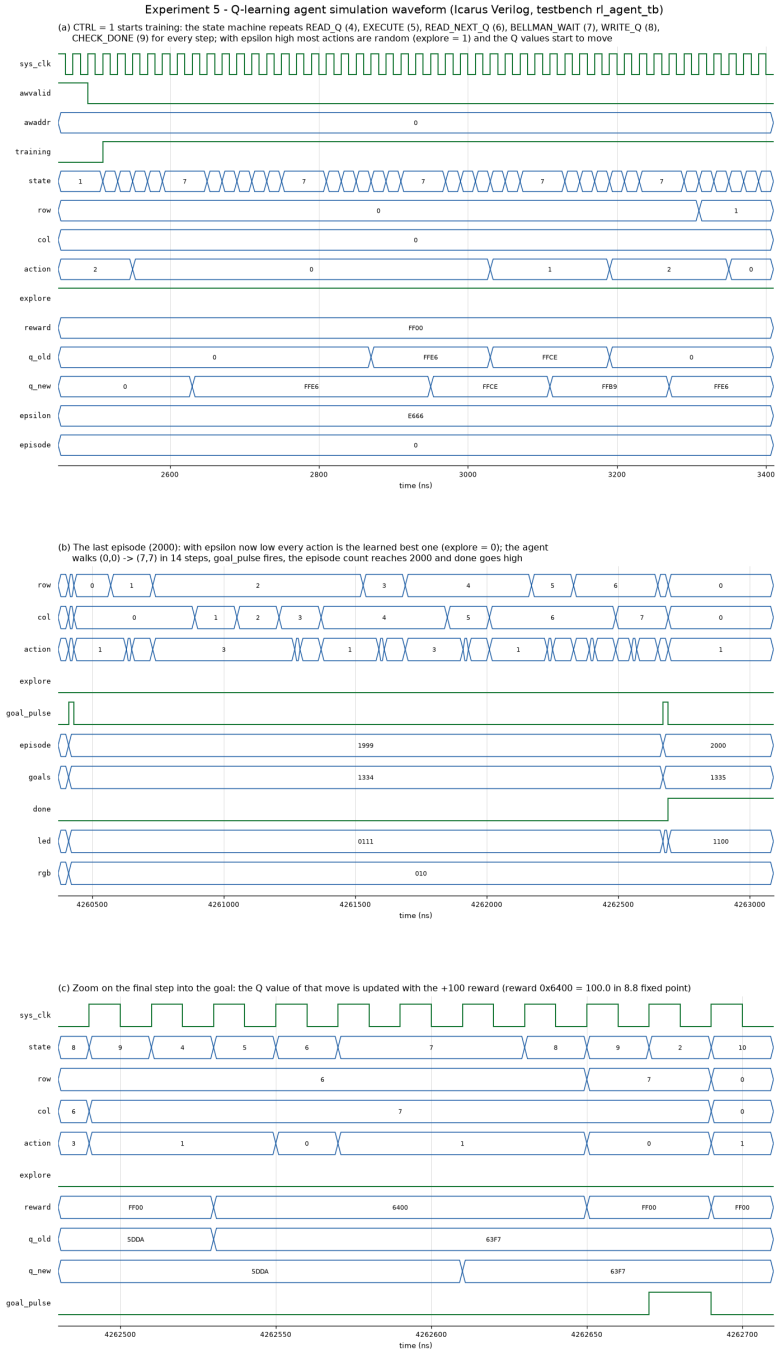


Figure 1 – Simulation waveform (Icarus Verilog, drawn from rl_wave.vcd)